Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/15/2007
PI Name:	Tai, Yu-Chong Ph.D.		
Project Title:	Handheld Body-Fluid Analysis System for Astronaut Health Monitoring		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ho, Chih-Ming (University of California, Los Angeles) Kasdan, Harvey (IRIS International)		
Grant/Contract No.:	NCC 9-58-TD00406		
Performance Goal No.:			
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Task Description:	(1) The specific aim of this project is to focus on the development and demonstration of an automated handheld blood count instrument, which is easy for astronaut to operate. A monolithic chip will be developed to perform measurements of red blood cell (RBC) count, mean cell volume (MCV), hematocrit, white blood cell (WBC) count and at least 2-part differential (lymphocytes verse granulocytes). The system should analyze a minimum of "1,000 RBC" and "200 WBCs" which corresponding to processing sample volume ~50-100 nL of whole blood.. The sample volume should be precisely measured ± 5%. (2) During the previous funding year, we made progress in electrical impedance sensing. We solved a serious problem for micro electrical impedance sensors, which is the large double layer impedance of electrode-electrolyte interface. The small surface area increases the double layer impedance and lowers the sensitivity. By either electroplating Platinum black on electrode surface or introducing an inductor parallel to the system to induce resonance, we achieve high signal-to-noise ratio when particles including blood cells passing the electrodes. At low AC frequency, the height of the signal was confirmed to be determined by the volume of the particles. The histograms of the signal height for whole blood and leukocyte rich plasma matched very well to previous published volume distributions of erythrocytes and leukocytes. Also in this year, we successfully demonstrated fluorescent sensing and counting for WBC count and 2-part differential. Cell nucleus fluorescent dye acridine orange stains DNA to be green and RNA to be red. As a result, non-lymphocytes have a higher red fluorescence level than lymphocytes due to their more active gene expression. So the red fluorescence channel was used for 2-part WBC differential, while the green fluorescent channel was used for WBC count due to its better signal to noise ratio. The unique feature of this technique is that whole blood can be used without dilution, which will simplify system design, device operation and buffer storage greatly. The concept was proved by constructing an optical system on optical bench with Ag laser as illumination source and also on a modified microscope with LED as illumination source. In both cases, high sensitive photon multiplication tubes (PMTs) were used as detectors. The systems can achieve throughput of 1000 WBCs per second, which means a WBC count can be finished in one second, which makes blood cell sedimentation not a problem any more. Finally, expanding our previous work on blood cell separation, we demonstrated separation of subtypes of lymphocytes from the rest of the blood cells. More specifically, we successfully achieved CD4+ T helper cell separation from the rest of the blood cells. The separation was built on our previous size based micro separators. The specificity was achieved by attaching monoclonal CD4+ antibody to beads, thus capturing the CD4+ T helper cells by immuno recognition. The beads with capture cells have their sizes changed so that they could be separation from unattached cells and bare beads by size. (3) The findings above are key components to meet the specific aims of the proposal. Progress in electrical impedance sensing enables us to sense and count RBCs and WBCs with high signal-to-noise ratio, while separation of RBCs and WBCs makes the electrical impedance sensing more sensitive by applying different geometries to RBCs and WBCs. We also showed under low AC frequency excitation, the signal height of the electrical impedance sensing was determined by the volume of the particle. So with proper calibration, the MCV could be measured directly by measuring individual cell volume and averaging them. Hematocrit can be determined indirectly by adding volumes of individual cells together. Optical fluorescent sensing demonstrated 2-part WBC differential from undiluted blood. Since fluorescent dye AO and anticoagulant EDTA can be coated on channel surface, so sample preparation will be minimized, which will be very convenient for space use. (4) Based on this year's progress, the next year will be focused on system integration. We propose two integrated devices. One separates RBCs and WBCs from diluted blood and RBCs and WBCs will be counted at different outlet channels with optimized geometries. This device will measure RBC count, WBC count, MCV and hematocrit. The other device use fluorescent detection on undiluted blood for WBC 2-part differential. The system will be miniaturized inside a small box using LED as illumination and PMTs as detectors. Various absolute volume measurement schemes will be experimented.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology developed from this project can be used in point-of-care situations. For example, it can find applications in emergency room, on ambulance and at home. The research approach is a combination of engineering, medicine and biology. The unique research experience gained from the project benefits the graduate students to become the next generation bioengineer.
Task Progress:	According to the original proposal, for the third year, we are supposed to finish research in on-chip aspiration and on-chip dilution, as well as RBC MCV. We are supposed to continue the work on WBC differential and start working on hematocrit and instrument integration. We are on the schedule. 1. on-chip aspiration: We decided to use miniaturized mini pumps for fluidic manipulation. We purchased P625/10K.015 peristaltic pump (Instech Laboratories, Inc., Plymouth Meeting, PA, USA). Initial testing confirmed that the operation of the pump meets our requirements. 2. on-chip dilution: For fluorescent detection, we demonstrated that dilution is not needed. For separation and then electrical impedance counting, we demonstrate that separation itself is an effective dilution. 3. RBC MCV: After we solved the large double layer impedance problem inherent to micro impedance sensing, high signal-to-noise ratio signals enabled us to perform statistical analysis on the signal height. We confirmed that at low AC excitation frequency (under 100kHz), the signal height is determined by the volume of the particle. Thus, individual volume of RBC can be determined and MCV can be calculated by averaging. 4. WBC differential: We demonstrated WBC differential by AO staining of WBCs and fluorescent detection at red channel. We are investigating the possibilities of 3-part and 5-part WBC differential by AO staining, which are beyond the original proposed. 5. Hematocrit: Hematocrit is supposed to be calculated by adding measured individual cell volume. 6. Instrument integration: We are in the progress of instrument integration. For fluorescent detection system, we moved from optical bench setup and modified fluorescent microscope to building a system much smaller that a shoe box. The first design has been made and we are in the process of implementation. Integrating cell separation and electrical impedance sensing is also under way.

Bibliography Type:	Description: (Last Updated: 08/30/2018)
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